Campus Laundry

Students Campus Laundry Booking and Management System

Installation and User Manual

How to install and run the system on a computer, and how students and administrators use it.

Version 1.0.0 · 4 October 2026 · Asemahle Sokaba · Avuziwe Mcata

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1. About the system

Campus Laundry lets students book a washer or dryer for a time slot instead of queuing. Students get a reminder before their slot and a notification when their laundry is done, can join a waiting list when everything is booked, read laundry tips and send feedback. The laundry administrator manages the laundry rooms and machines, answers feedback and sees booking history and usage reports.

Who	Uses the system to	Read
Student	Book, start and collect laundry; waiting list; notifications; tips; feedback	Part B (Sections 5–11)
Administrator	Set up rooms and machines; mark machines out of order; reply to feedback; reports; manage accounts	Part C (Sections 12–17)
Whoever installs it	Install and run the system on a computer	Part A (Sections 2–4) and Section 18

How a laundry booking works

1. Book a washer or dryer for a slot (today or up to 7 days ahead).
2. 30 minutes before the slot you get a **reminder** (🔔).
3. At the laundry room, load the machine and tap **Start cycle** (from 5 minutes before your slot until 15 minutes after it starts).
4. When the machine's cycle time has passed you get **"your laundry is done"**.
5. Take your laundry out and tap **Collected**. The next student booked on that machine is told it is free.

> **Important:** If you don't start the cycle within 15 minutes of your slot starting, the booking is marked as **missed** and the machine is released for others.

Part A · Installation

The system runs on one computer (for example the residence office computer). That computer runs **MySQL** (the database) and the **Campus Laundry server** (Node.js). Students and staff use it in a web browser: on the same computer at http://localhost:3003, or from phones and laptops on the same Wi-Fi (Section 18.2).

Step	Task	Time
1	Copy the project folder to the computer	5 min
2	Install Node.js	5 min
3	Install and configure MySQL Server	10–15 min
4	Open a terminal in the project folder	1 min
5	Create the settings file (.env)	5 min
6	Install the project's packages	1–3 min
7	Create the database	1 min
8	Create the first administrator	1 min
9	Start the system and log in	2 min
10	(Optional) Check the installation	5 min

> **Tip:** You need an internet connection while installing (steps 2, 3 and 6). After that, the system works without internet.

2. What you need

2.1 Computer

- Windows 10 or 11 (64-bit). macOS and Linux also work; see Appendix B.
- At least 4 GB of RAM and 1 GB of free disk space.
- An account that is allowed to install software (administrator rights).

2.2 Software to install

Software	Version	What it is for	Where to get it
Node.js (includes npm)	24 LTS (tested with 24.19.0)	Runs the server; npm installs its packages	https://nodejs.org (choose "LTS")
MySQL Community Server	8.4 LTS (tested with 8.4.9)	The database	https://dev.mysql.com/downloads/mysql/ (choose 8.4 LTS)
A web browser	Any current version	Using the system	Microsoft Edge, Google Chrome or Firefox
Microsoft Edge, Word (optional)	Current	Only for the browser tests and rebuilding the documents	Not needed to run the system

2.3 What is in the project folder

Item	What it is	Copy it?
server.js, routes/, lib/, public/	The application	Yes
db/	Database creation script and migrations	Yes
tools/	Admin commands, tests and sample data	Yes
docs/	This manual, the test report and screenshots	Yes
package.json, package-lock.json	The list of packages needed	Yes
.env.example	Template for the settings file	Yes
.env	Settings for the OLD computer (database password, secret key)	No: create a new one (step 5)
node_modules/	Downloaded packages (large)	No: re-created in step 6
.claude/	Editor settings from development	No

> **Important:** The database is not inside the project folder: accounts and bookings live in MySQL. A new installation starts with an empty database. To bring existing data across, see Section 19.

3. Installation steps

Step 1: Copy the project folder

1. Copy the whole **laundry** folder, apart from **node_modules**, **.claude** and **.env**, using a USB drive, a zip file or a cloud drive.
2. On the new computer, put it somewhere simple, for example **C:\CampusLaundry\laundry**.

> **Tip:** Avoid keeping the folder in OneDrive, Dropbox or Google Drive on the new computer. Syncing thousands of package files makes installing and starting much slower.

Step 2: Install Node.js

Option A: installer (easiest). Go to https://nodejs.org, download the **LTS** Windows installer (.msi), run it and keep all the default options.

Option B: command. In Windows Terminal or PowerShell:

```
winget install --id OpenJS.NodeJS.LTS
```

Check it worked. Close every terminal window, open a **new** one, and run:

```
node -v
npm -v
```

> **Important:** If you see "npm is not recognized", the terminal was opened before Node.js was installed. Close all terminal windows (and editors such as VS Code) and open a new one.

Step 3: Install and configure MySQL Server

Option A: installer. Go to https://dev.mysql.com/downloads/mysql/, choose **8.4 LTS**, and download the **Windows (x86, 64-bit), MSI Installer** ("No thanks, just start my download"). Run it and choose **Typical**.

Option B: command.

```
winget install --id Oracle.MySQL
```

Configure the server. When installation finishes, **MySQL Configurator** opens. If it doesn't, search the Start menu for "MySQL Configurator".

1. Config type: **Development Computer**. Port: **3306** (keep the default).
2. Set a **root password** and write it down; you need it in step 5.
3. Keep **Configure MySQL Server as a Windows Service** and **Start the MySQL Server at System Startup** ticked. The service is usually called **MySQL84**.
4. Click **Next** through the remaining pages, then **Execute**, then **Finish**.

Check it is running: open the **Services** app and look for **MySQL84** with status **Running**, or in PowerShell:

```
Get-Service MySQL84
```

> **Tip:** If this computer already runs another project with MySQL 8.4, skip this step: they can share the same MySQL server because each uses its own database.

Step 4: Open a terminal in the project folder

In File Explorer, open the **laundry** folder, right-click an empty area and choose **Open in Terminal**, or type:

```
cd "C:\CampusLaundry\laundry"
```

Step 5: Create the settings file (.env)

```
copy .env.example .env
notepad .env
```

Setting	What to put
DB_HOST, DB_PORT, DB_USER	Leave as `localhost`, `3306`, `root`
DB_PASSWORD	The MySQL root password from step 3
DB_NAME	Leave as `campus_laundry`
PORT	Leave as `3003` (change only if another program uses it)
SESSION_SECRET	A long random text that keeps logins secure. Create one with the command below and paste it in.

```
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

> **Important:** .env contains passwords. Do not share it or copy it to other computers.

Step 6: Install the project's packages

```
npm install --omit=dev
```

This downloads what the system needs into a new **node_modules** folder and ends with a line such as "added 80 packages".

> **Tip:** `--omit=dev` skips the tools that are only used for browser tests, sample data and rebuilding the documents. To include them, run `npm install` instead.

Step 7: Create the database

```
npm run db:setup
```

You should see:

```
Applying migration 001_initial_schema…
Database "campus_laundry" is up to date.
There is no administrator yet. Create one with:  npm run admin:create -- "Full
Name" email@example.com
```

This creates an empty database with all 9 tables and 8 starting laundry tips. It is safe to run again at any time: it never deletes data.

Step 8: Create the first administrator

```
npm run admin:create -- "Your Name" you@example.com
```

It prints a **temporary password**. Write it down; you will choose your own password when you log in.

Step 9: Start the system

```
npm start
```

When you see **Campus Laundry running at http://localhost:3003**, open a browser and go to:

```
http://localhost:3003
```

Log in with the administrator email and temporary password, then choose your own password. Keep the terminal window open while the system is in use: it also sends the reminders and "laundry done" notifications. Press **Ctrl + C** in it to stop.

Step 10 (optional): Check the installation

These checks create their own temporary test data and remove it again, so they are safe on a real installation.

```
npm run db:check
```

Expect "MATCH" and "43/43 integrity rules enforced". Then, **while the system is running**, open a second terminal in the project folder and run:

```
npm run test:api
```

Expect "84/84 API tests passed. All test accounts and data were removed." With the full `npm install`, `npm test` also runs the browser and load tests (see the Test Report).

4. First-time setup (administrator)

1. Go to **Machines & rooms → Laundry rooms** and add each laundry room: its name, location, opening and closing times and the **slot length** (for example 60 minutes). Students book one slot at a time.
2. Go to **Machines** and add every washer and dryer: the room, a short **code** written on the machine (e.g. W1, D2), the type and its **cycle time** in minutes. The cycle must fit in one slot.
3. Check the **Laundry tips** and edit them if you like.
4. Tell students the address of the system so they can **register**.

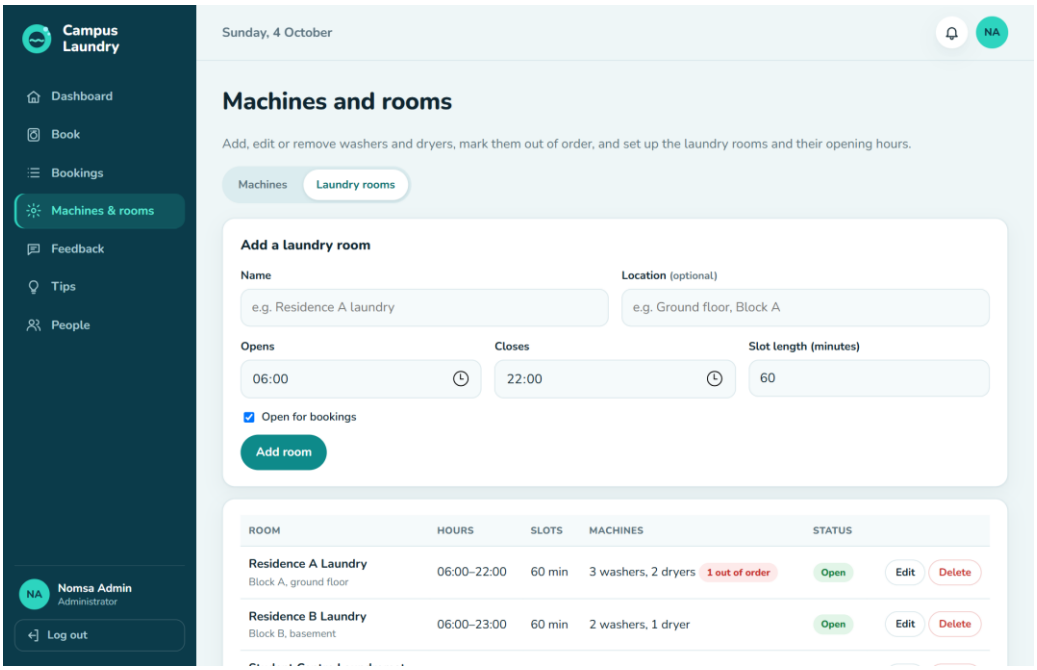


Figure 1: Machines & rooms → Laundry rooms: opening hours and slot length

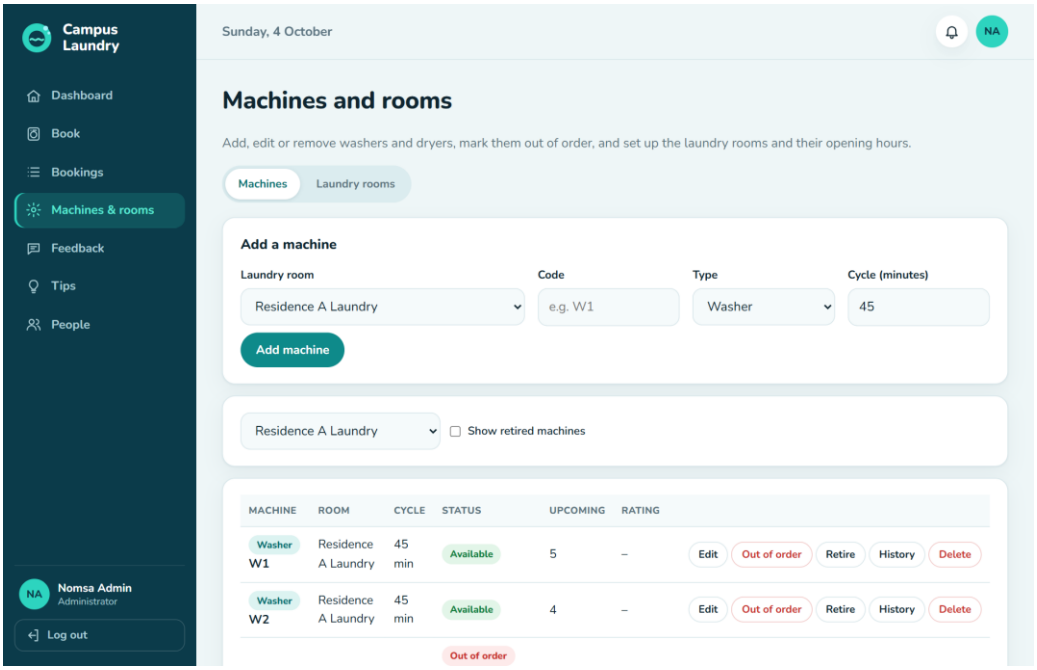


Figure 2: Machines: add washers and dryers; mark them out of order or retire them

Part B · Student guide

5. Getting started

5.1 Register

1. Open the system and click **Register**.
2. Enter your full name, **student number** (8–10 digits), email and a password of at least 8 characters.
3. Choose **your laundry room** (it opens first when you book) and, if you like, your phone number.
4. Click **Create account**. You are logged in and taken to **My laundry**.

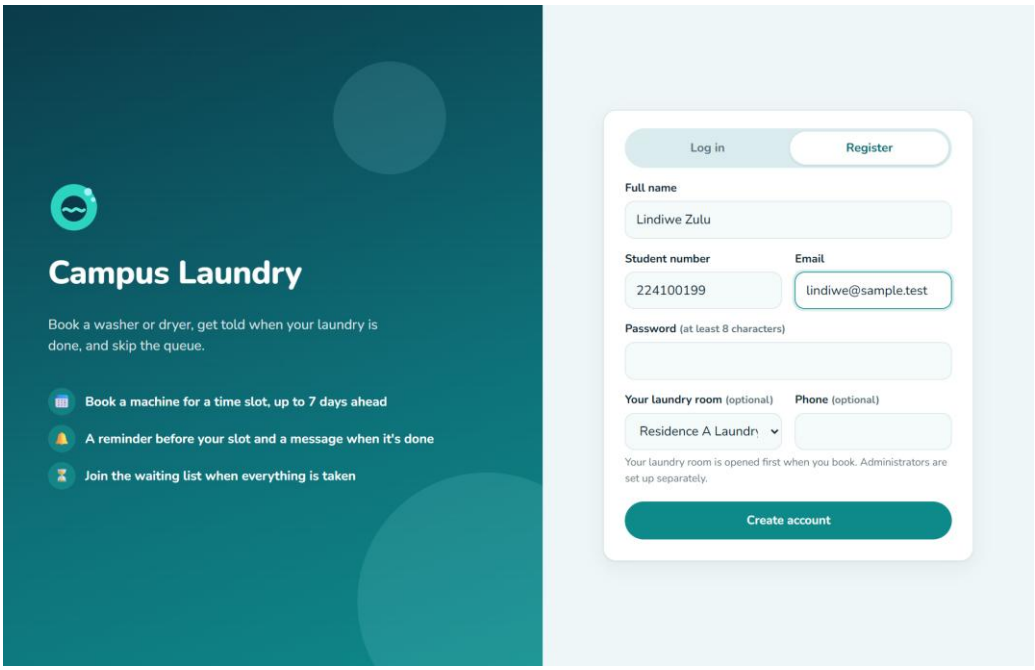


Figure 3: Registering as a student

5.2 Log in

Log in with your **email or your student number** and your password. Forgot your password? Ask the laundry administrator: they can give you a temporary password, and you then choose a new one.

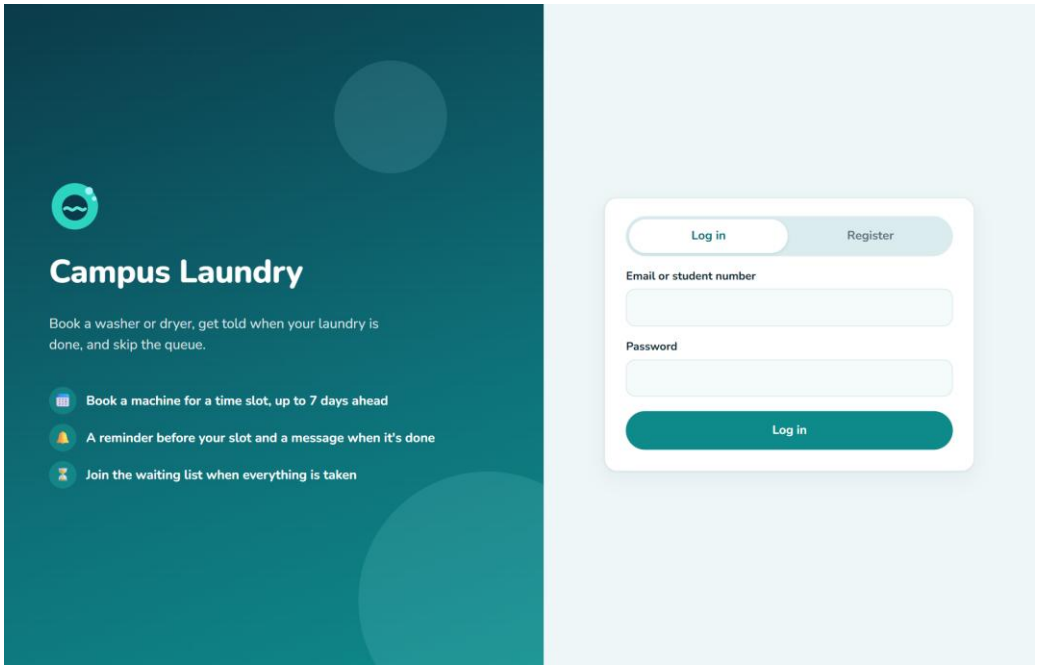


Figure 4: Log in

5.3 The menu and the bell

The menu is on the left of the screen. On a phone it is a bar along the bottom of the screen with short names (Mine, Book, Tips, Feedback). The bar at the top shows today's date, the bell and your initials.

Menu	What it is for
My laundry	Your bookings: start the cycle, mark it collected, cancel or reschedule; your waiting list; your history
Book a machine	Find a free washer or dryer and book it
Laundry tips	Advice on sorting, detergents, fabrics and machine care
Feedback	Rate the service, comment, or report a problem
Bell (top right)	Your notifications; the red number shows how many are unread
Your initials (top right) or your name (bottom of the menu)	Your profile: details, laundry room, phone, password
Log out (bottom of the menu; the arrow icon at the top right on a phone)	Log out

5.4 Your profile

Click your initials at the top right (or your name at the bottom of the menu) to change your name, email, laundry room or phone, change your password, or delete your account. Your student number cannot be changed.

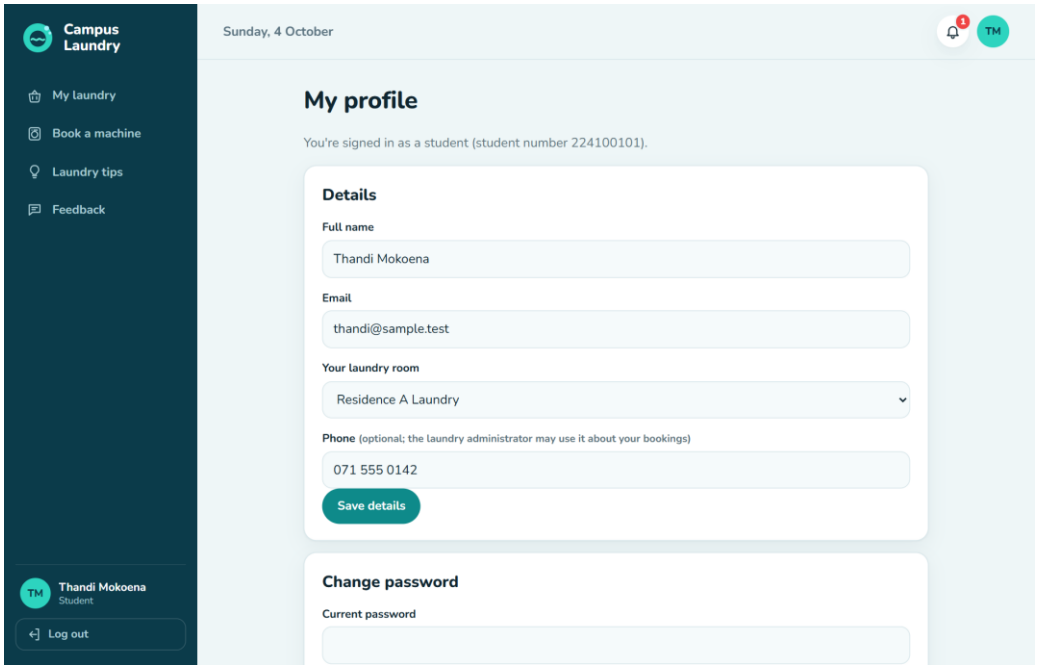


Figure 5: My profile

6. Booking a machine

1. Click **Book a machine**.
2. Choose the **laundry room** at the top right, and the **day** (today or up to 7 days ahead).
3. The grid shows every machine (rows) and every time slot (columns). Scroll sideways to see later slots.
4. Click a green **Book** slot. A bar appears at the bottom with the machine, day and time.
5. Click **Confirm booking**. You are taken to **My laundry** with your new booking.

In the grid	Means
Book (green)	Free: click to book
Yours (blue)	Your booking
Taken (grey)	Another student booked it
Out (red stripes)	The machine is out of order; the reason is shown under the machine
Empty	The slot has passed

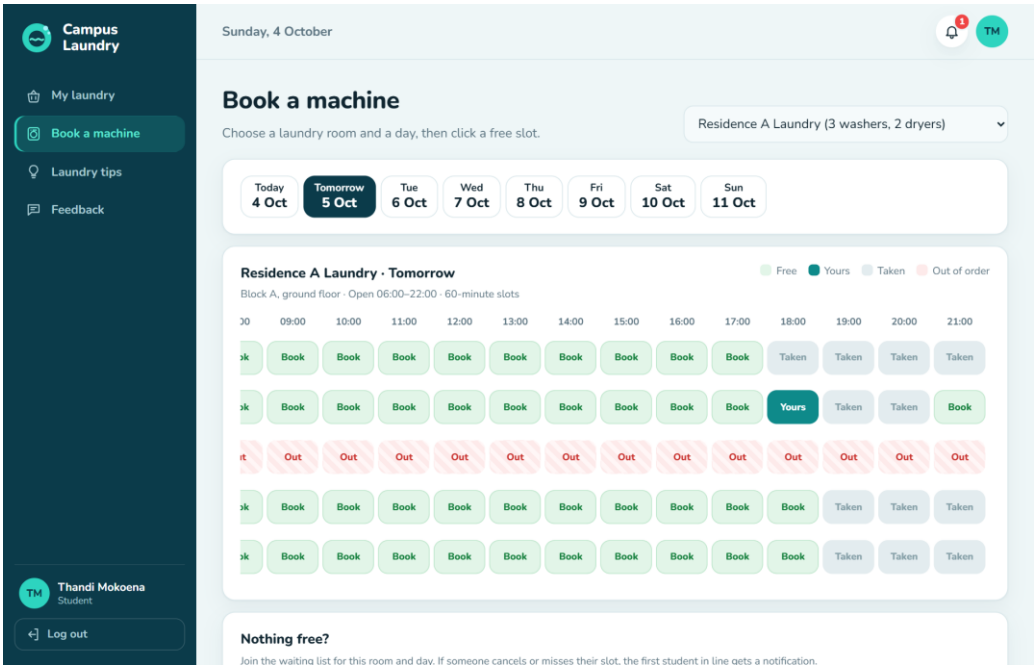


Figure 6: The booking grid: free, taken and out-of-order slots

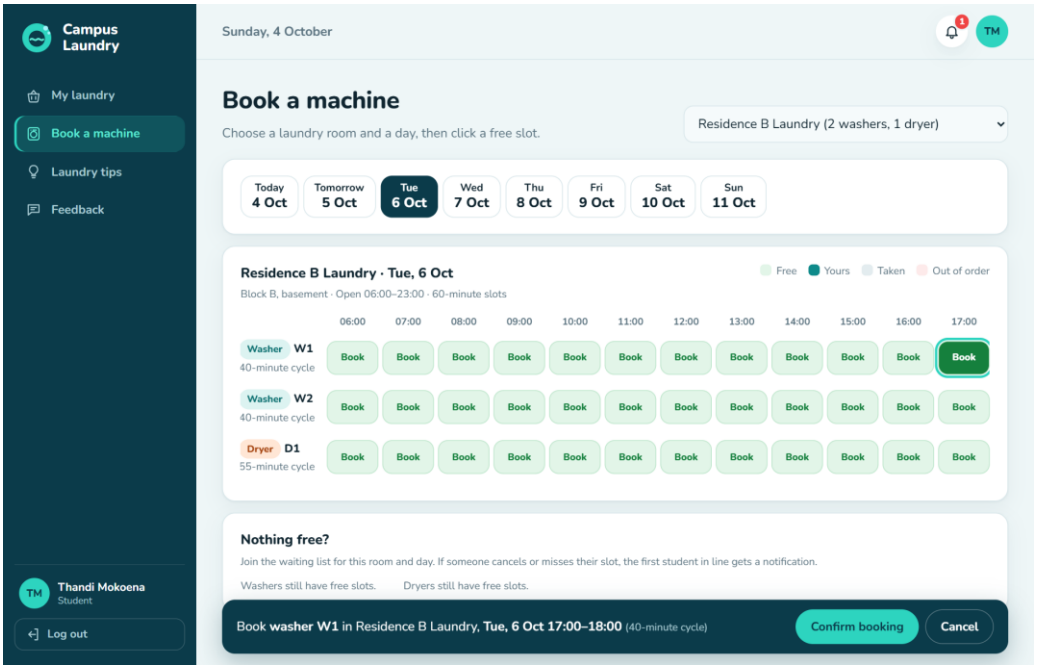


Figure 7: Confirming a booking

6.1 Booking rules

- You can have at most **3 open bookings** at a time.
- You can't book two machines for the same time.
- If someone books the same slot a moment before you, you'll see "Sorry, someone has just booked that slot". Choose another one.

6.2 Reschedule or cancel

On **My laundry**, click **Reschedule** to pick a new slot in the grid (then **Move booking**), or **Cancel** to free the slot for someone else. Both are possible until your slot starts.

7. Doing your laundry

1. Go to the laundry room in time for your slot and load the machine.
2. Open **My laundry** and tap ▶ **Start cycle**. You can do this from 5 minutes before your slot until 15 minutes after it starts.
3. A progress bar shows how much time is left.
4. When the cycle is done you get a notification. Take your laundry out promptly.
5. Tap ✓ **Collected**. The machine is now free, and the next student booked on it is told.

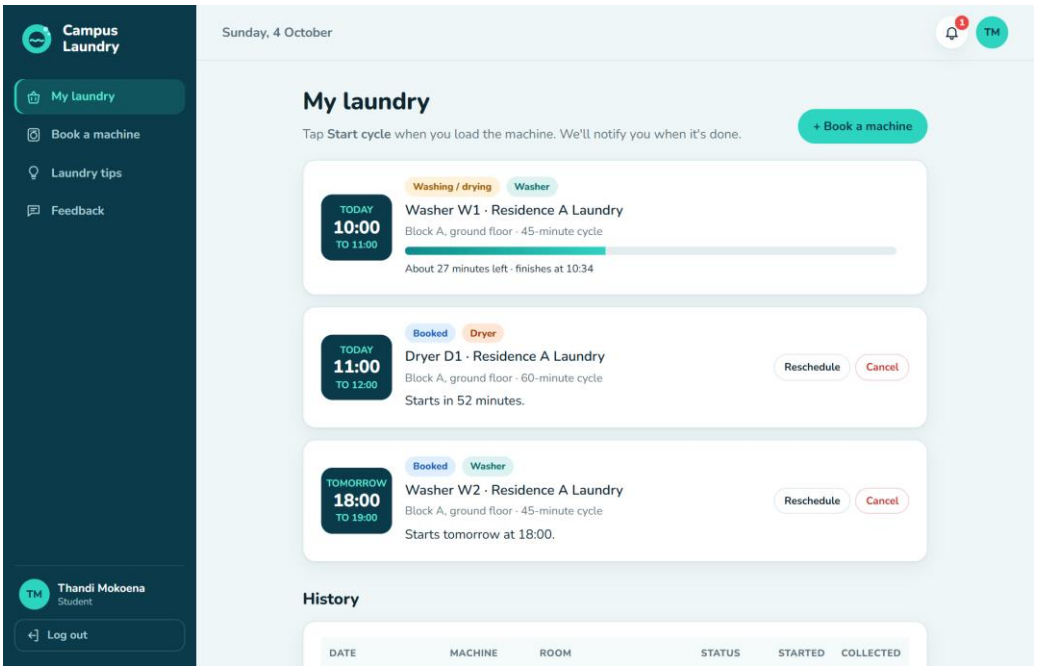


Figure 8: My laundry: a cycle running, a booking ready to start, and an upcoming booking

> **Important:** Not started within 15 minutes? The booking is marked as **Missed** and the machine is released. Missed bookings show up in your history and in the administrator's reports.

8. The waiting list

When every washer (or dryer) in a room is booked for a day, the **Book a machine** page shows **Join the waiting list for a washer / dryer**. If someone cancels or misses their slot, the **first** student in line gets a notification and can book it. Your waiting-list places are listed on **My laundry**, with how many students are ahead of you. Booking a machine of that type removes you from the list automatically.

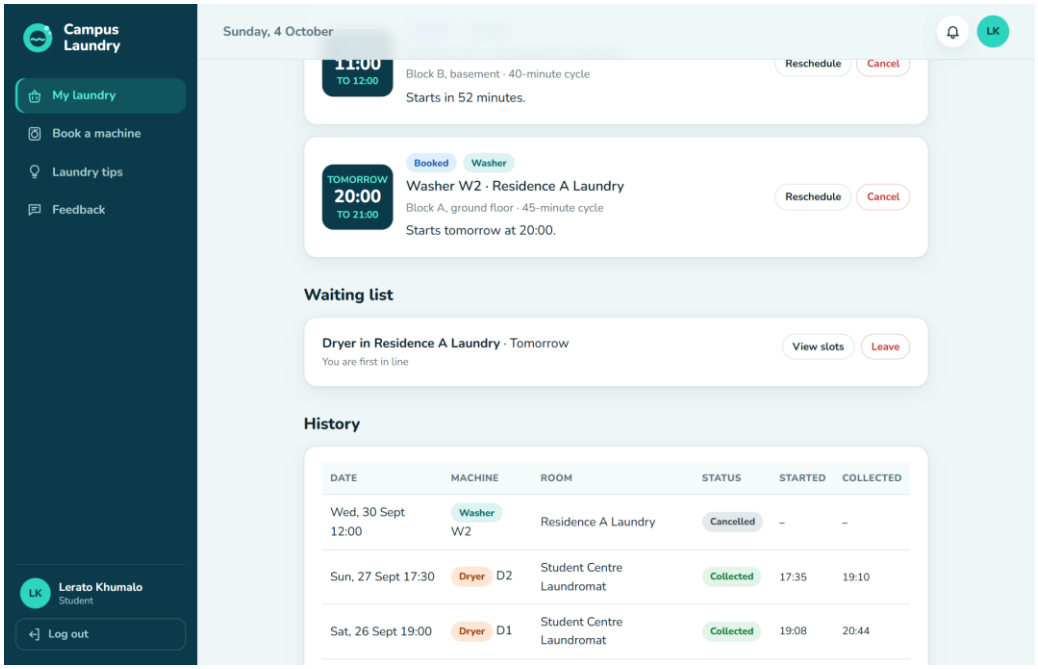


Figure 9: My laundry: on the waiting list for a dryer

9. Notifications

Click the **bell** to see your notifications. **Open** takes you to the right page; **Mark read** or **Mark all as read** clear the red number.

Notification	When
⏰ Reminder	30 minutes before your slot
✅ Laundry done	When your cycle has finished
🟢 Machine free	The machine you booked later today is free now, or a machine became free for your waiting list
⚠️ Booking cancelled	The administrator cancelled your booking, for example because the machine is out of order
⌛ Missed	You did not start the cycle within 15 minutes
💬 Feedback reply	The administrator answered your feedback

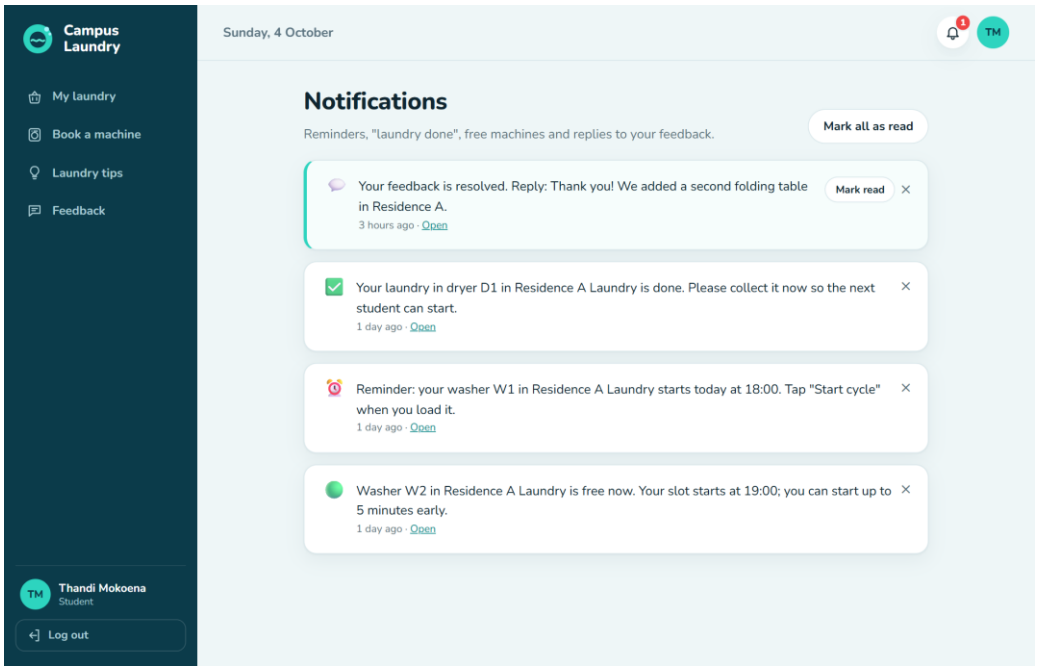


Figure 10: Notifications

10. Laundry tips

Laundry tips gives advice on **sorting clothes**, **choosing detergents**, **fabric types** and **machine care**. Click a category to see only those tips.

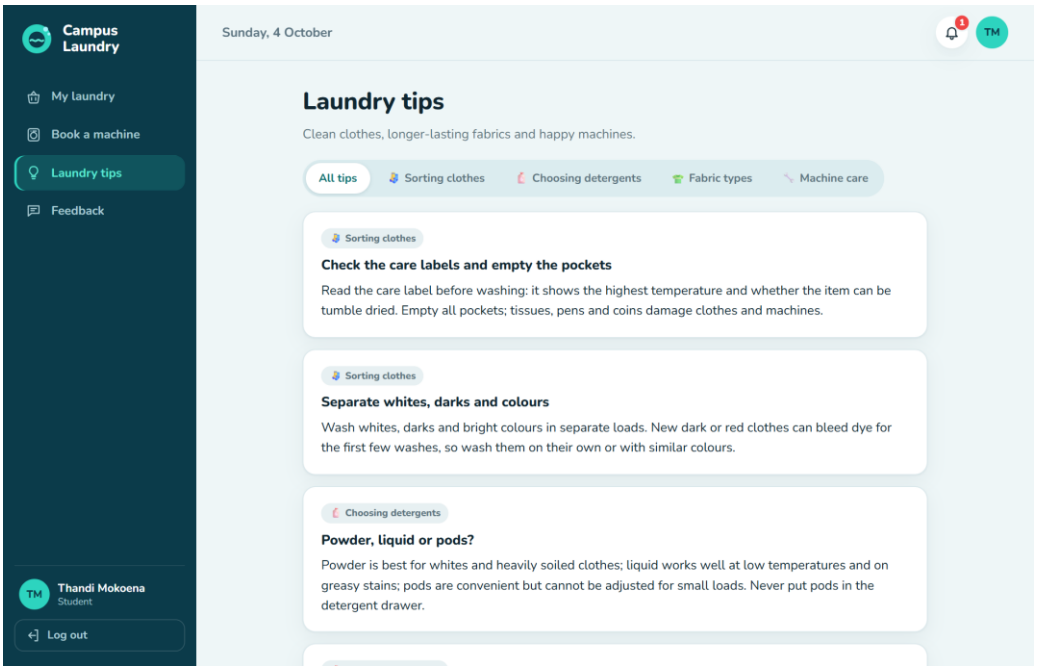


Figure 11: Laundry tips

11. Feedback

1. Click **Feedback** and choose **Rate**, **Comment** or **Report an issue**.
2. Choose the laundry room and, if it is about one machine, the machine.
3. For a rating, click 1 to 5 stars. Add a comment (required for comments and issues).
4. Click **Send feedback**. Your feedback and the administrator's replies appear under **My feedback**, and you get a notification when they reply.

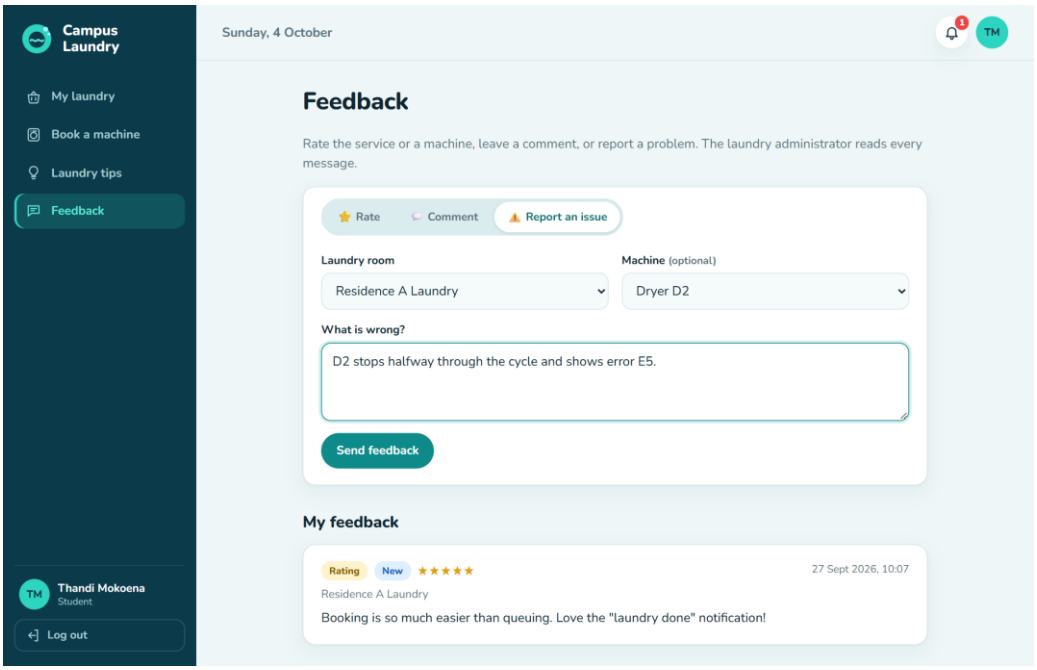


Figure 12: Reporting an issue with a machine

Part C · Administrator guide

12. Dashboard

The **Dashboard** opens when you log in. **Right now** shows today's bookings, machines running, loads waiting to be collected, machines out of order, open issues and students on waiting lists. The **Usage report** (last 7, 30 or 90 days) shows bookings, loads washed, cancellations, the no-show rate and the average rating, with bookings per day, the busiest hours, use per machine and per room.

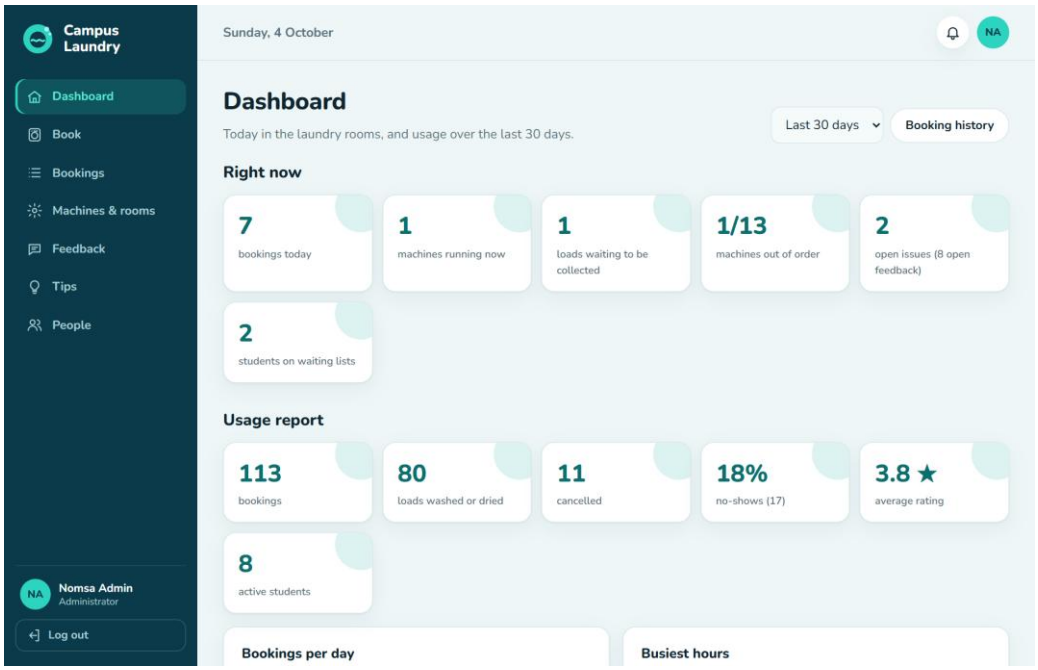


Figure 13: Dashboard and usage report

13. Machines and rooms

Task	How
Add a machine	Machines & rooms → fill in room, code, type and cycle time → **Add machine**
Edit a machine	**Edit** in its row; change the details → **Save changes**
Out of order	**Out of order** → type the reason. The machine can't be booked; its upcoming bookings are cancelled and those students are notified with the reason.
Repaired	**Back in service**. The **History** button lists every past outage.
Remove a machine	**Delete** works only for a machine that was never booked. Otherwise use **Retire**: it disappears from the booking grid but its history stays (its upcoming bookings are cancelled).
Rooms	Laundry rooms tab: add or edit rooms and opening hours. Opening hours and slot length can only change when the room has no upcoming bookings. Untick **Open for bookings** to close a room. A room with machines can't be deleted.

14. Feedback and issues

Feedback lists open feedback first (issues before ratings and comments). For each one you can write a **reply**, set the **status** (New, Being looked at, Resolved, Closed) and **Save**; the student is notified. Closing (dismissing) feedback needs a reply explaining why. For an issue about a machine, **Mark machine out of order** takes it out of service in one step.

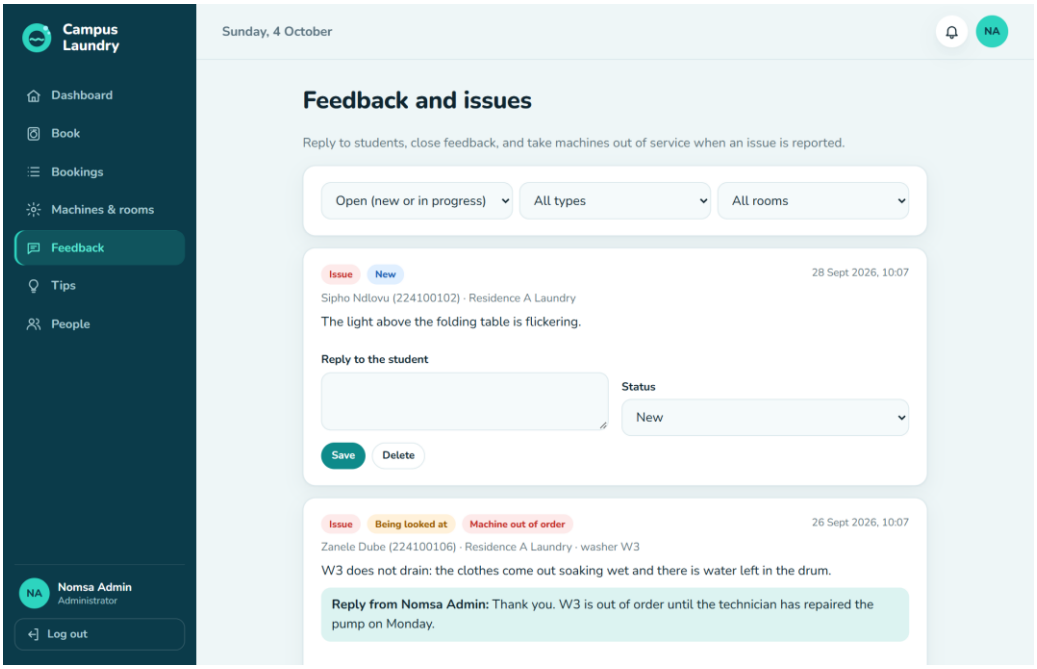


Figure 14: Feedback and issues

15. Booking history

Bookings lists every booking in every room. Filter by student (name, student number or email), dates, room and status. You can **Cancel** an upcoming booking; the student is notified.

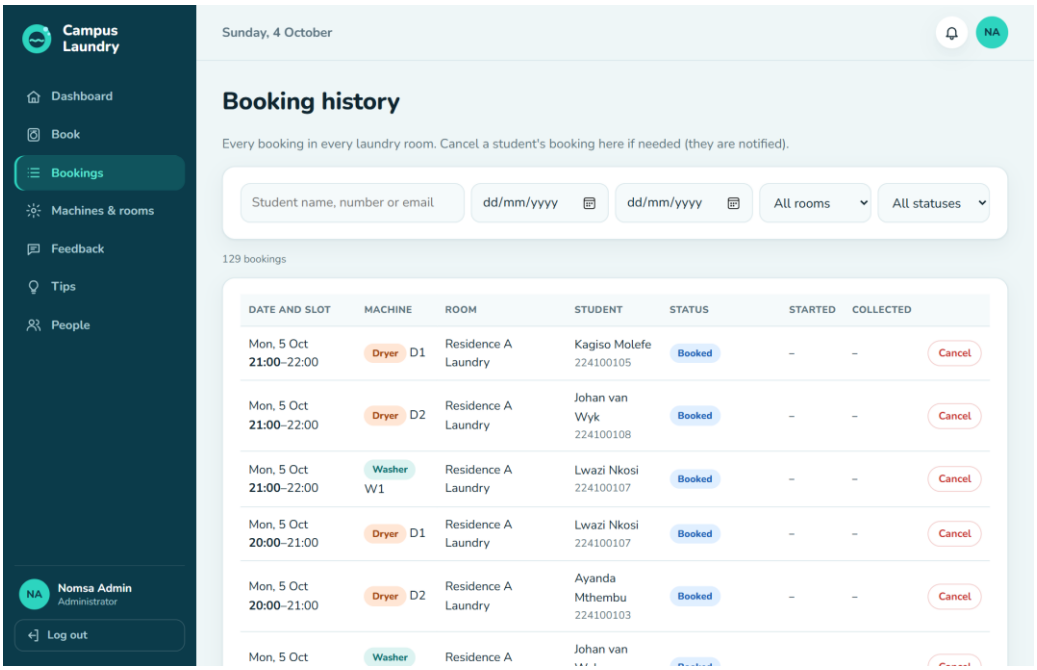


Figure 15: Booking history

16. People

- Search by name, email or student number.
- Change a role to **Admin** to make another administrator.
- **Deactivate** someone who should no longer use the system: they can't log in and their upcoming bookings are cancelled; their history stays. **Reactivate** reverses it.
- **Reset password** shows a temporary password once; they choose a new one when they log in.
- **Delete** removes the account; their past bookings and feedback stay in the reports as "former student".

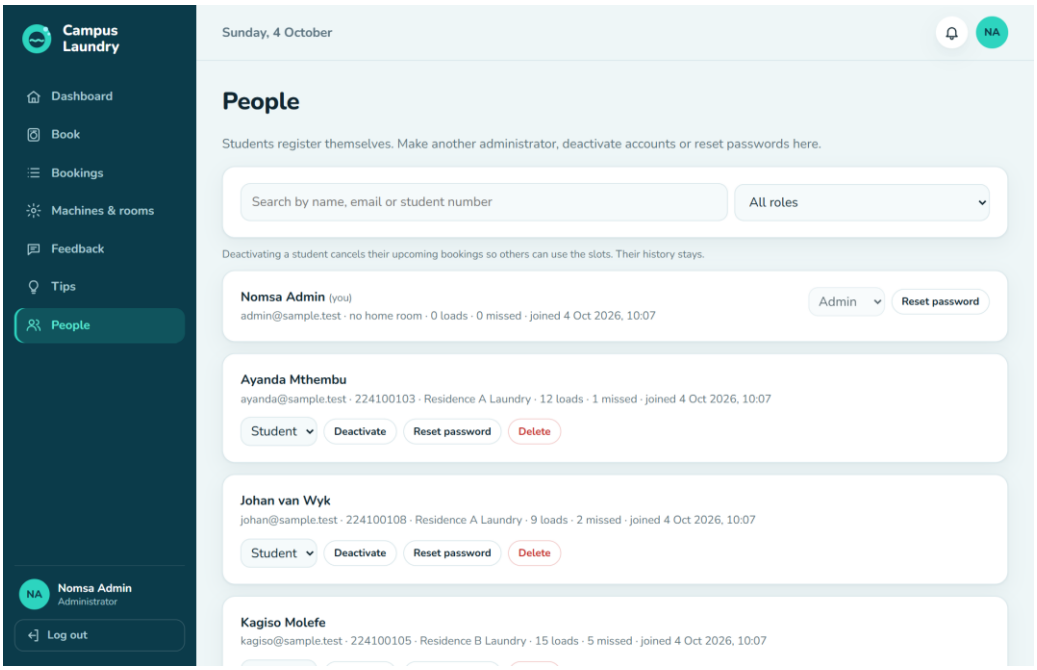


Figure 16: People

17. Laundry tips (administrator)

On **Tips**, click **+ New tip**, choose the category, write a title and the tip, and **Save tip**. Untick **Published** to keep it as a draft that students can't see. **Edit** and **Delete** are on each tip.

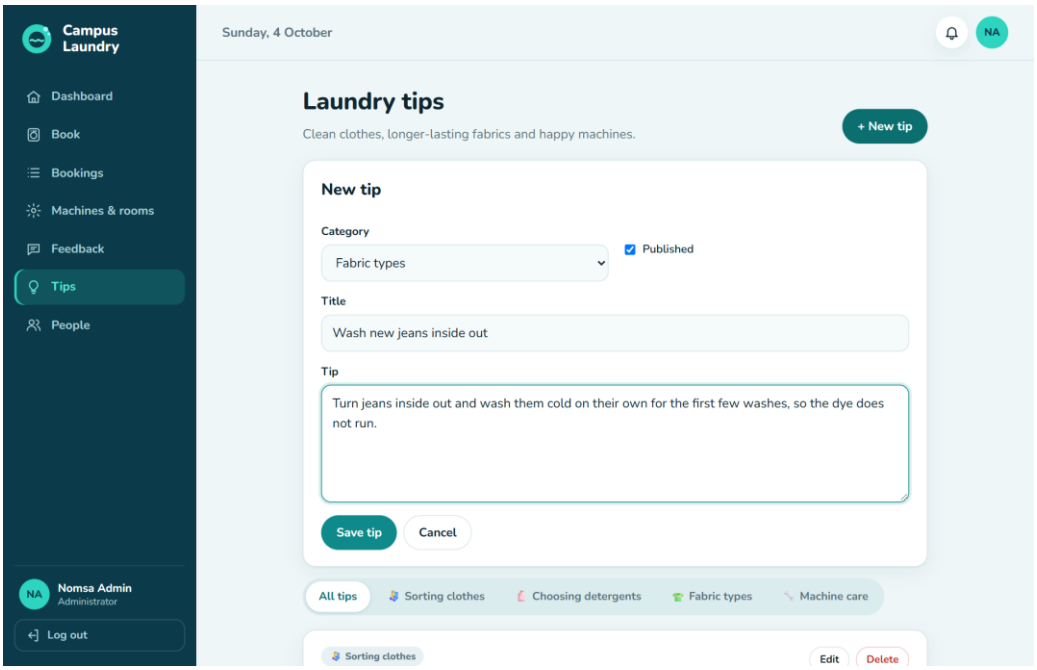


Figure 17: Adding a laundry tip

18. Everyday use

18.1 Starting and stopping

- MySQL starts automatically with Windows.
- Start the system: open a terminal in the project folder and run `npm start`.
- Stop it: press **Ctrl + C** in that terminal, or close the window. While it is stopped, no reminders or "laundry done" notifications are sent; they catch up when it starts again.

18.2 Using it from phones

1. On the computer running the system, run `ipconfig` and note the **IPv4 Address**, e.g. `192.168.1.25`.
2. The first time it starts, Windows may ask whether to allow Node.js through the firewall. Allow it on **Private networks**.
3. On phones on the same Wi-Fi, open `http://192.168.1.25:3003` (use your own address). The pages are designed for phone screens.

> **Important:** The system uses plain http, which is fine on a trusted local network. Do not put it directly on the internet without setting up https.

18.3 Backing up

Back up regularly. Open **Command Prompt**, go to the project folder and run (it asks for the MySQL root password):

```
cd /d "C:\CampusLaundry\laundry"
"C:\Program Files\MySQL\MySQL Server 8.4\bin\mysqldump.exe" -u root -p --
databases campus_laundry --result-file=laundry-backup.sql
```

18.4 Demonstration data

For a presentation you can fill the system with fictional data: three laundry rooms, students with @sample.test emails and two weeks of bookings. It needs the full `npm install` and the system running:

```
npm run sample:seed
```

Log in as **admin@sample.test** or a student such as **thandi@sample.test**; the password is in `tools/sample-data.js`. Remove all of it again afterwards (only the sample data is deleted):

```
npm run sample:clean
```

19. Moving existing data to another computer

1. On the **old** computer, make a backup as in Section 18.3 and copy **laundry-backup.sql** to the new computer.
2. On the **new** computer, complete installation steps 1–6, but **skip steps 7 and 8**.
3. Put **laundry-backup.sql** in the project folder, open **Command Prompt** there and import it:

```
"C:\Program Files\MySQL\MySQL Server 8.4\bin\mysql.exe" -u root -p < laundry-backup.sql
```

4. Run `npm run db:setup` (it applies any newer updates).
5. Start with `npm start` and log in with the existing accounts.

20. Troubleshooting

Problem	Likely cause	What to do
"npm is not recognized"	The terminal was opened before Node.js was installed	Close all terminals (and VS Code) and open a new one.
"Cannot reach the database" or ECONNREFUSED	MySQL is not running	Services → MySQL84 → Start. Set its startup type to Automatic.
db:setup: "ER_ACCESS_DENIED_ERROR"	Wrong DB_PASSWORD in .env	Put the correct MySQL root password in .env.
npm start: "EADDRINUSE :::3003"	Another program is using port 3003	Close it, or set another PORT in .env and open that address.
The only administrator forgot their password	Nobody else can reset it in the app	Run `npm run admin:reset-password -- admin@email` and use the temporary password shown.
A student forgot their password		Administrator: People → Reset password.
"This account has been deactivated"	The administrator deactivated it	Administrator: People → Reactivate.
No rooms to book	No laundry rooms or machines yet, or the room is closed	Administrator: Machines & rooms (Section 4).
"You can have at most 3 open bookings"	The limit per student	Finish (Collected) or cancel a booking first.
"Start cycle" button missing	Too early, or more than 15 minutes into the slot	It appears 5 minutes before the slot; after 15 minutes the booking is released.
"The previous cycle is still running"	The student before you started late	Wait until it finishes, then start.
No reminders or "done" notifications	The server was stopped	Keep `npm start` running; notifications catch up when it restarts.
Changes to .env have no effect	The server reads .env only when it starts	Stop (Ctrl + C) and run `npm start` again.

Appendix A: Command reference

Command	What it does
npm install --omit=dev	Installs the packages the system needs
npm run db:setup	Creates the database or brings it up to date. Never deletes data.
npm run admin:create -- "Name" email	Creates an administrator (or promotes an existing account) with a temporary password
npm run admin:reset-password -- email	Gives anyone a temporary password
npm start	Starts the system at http://localhost:3003 (stop with Ctrl + C)
npm run db:check	Checks the database structure and 43 integrity rules (leaves no data)
npm run test:api	84 end-to-end tests while the system is running (leaves no data)
npm test	Every test suite, including browser and load tests (needs the full npm install)
npm run sample:seed / sample:clean	Adds / removes fictional demonstration data
npm run db:reset	**Deletes the whole database** and creates an empty one

Appendix B: macOS and Linux

Step	Windows	macOS / Linux
Install Node.js	Installer or winget	Installer from https://nodejs.org (macOS), or your package manager
Install MySQL 8.4	MSI installer + MySQL Configurator	DMG installer from dev.mysql.com (macOS), or your package manager
Copy the settings file	copy .env.example .env	cp .env.example .env
Edit the settings file	notepad .env	nano .env (or any text editor)

All the npm commands are the same on every system. The browser tests and document tools use Microsoft Edge and Word, so they run on Windows.